Controlled evidence methodology

Single-case pilot: 1 RepoPairBench scenario x 3 evidence conditions x 2 harnesses

1 Paired repository source

Scenario 24e78349 / iRedAdmin
Vulnerable + fixed snapshots: researcher inspection

2 Researcher validation

Scoped label, threat model and evidence dependency
Fixed source and answer key stay researcher-only

3 Deterministic condition construction

Unchanged vulnerable target; physical source copies; SHA-256 manifests; leakage and boundary checks

E1 Sufficient initially

Expanded initial source packet
Context also available in workspace
19 files / 22 initial items

E2 Retrievable context

Shared limited initial packet
Missing context can be read
19 files / 4 initial items

E3 Unavailable context

Same initial packet as E2
Renderer and templates omitted
13 files / 4 initial items

CONTROL A: E1/E2 source identical

CONTROL B: E2/E3 initial evidence identical

4 Fresh sessions in six experimental cells

Same frozen task instructions; Qwen Code 0.24.6 or OpenCode 1.18.30
Same qwen3.5:9b-q4_K_M via Ollama; 16,384-token configured context; read-only source tools

5 Preserve and normalize

Raw traces + answers; mechanical metrics

6 Researcher scoring

Recognition / acquisition / abstention / grounding

Six draft cells; eight recorded attempts retained. No replicated sampling or harness ranking.